Wonjong Peter Lee

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Computer Engineering junior with SoC/NPU verification experience.

Education

University of Illinois Urbana-Champaign — Champaign, IL
B.S., Computer Engineering — *Aug. 2021 – May 2027*

- GPA: 3.92/4.0, Dean's List {FA21, SP22, SP23}
- Relevant Coursework: Computer Architecture, Parallel Programming (CUDA), Computer Systems Engineering, Digital Systems Laboratory, Analog Signal Processing, Data Structures, Linear Algebra, Discrete Mathematics

Experience

Design Verification Intern — May. 2026 – Aug. 2026
Apple, Inc. — *Austin, TX*

Post-Silicon Validation Intern – Live Demo at Hot Chips Symposium — Apr. 2025 – Aug. 2025
Rebellions, Inc. — *Seongnam, Korea*

- Took ownership of Trace32-based post-silicon validation on core IPs of the (REBEL_Quad) big-chip NPU SoC.
- Built a LLVM-based translator to generate Trace32 scripts from C firmware, expanding validation beyond block-level IP to 1,000+ workload patterns *before* CP/FW/DRAM bring-up;
- Above project became the preferred early-silicon screening/sorting method in-house and at the design-house partner; authored user guide for partner use.
- Developed a register-programming GUI that accelerated on-silicon debug and failure tracking for bring-up teams.
- Contributed to *Hot Chips 2025* single chip Llama 3.3 70B live demo, delivered in ∼30 days of chip arrival.

The Republic of Korea Army – Sergeant — Sep. 2023 – Mar. 2025

Design Verification Intern - Special Projects — Mar. 2023 – Aug. 2023
Rivian Automotive, Inc. — *Champaign, IL*

- Designed a Python DV infrastructure for DUT simulation with live performance telemetry and a Grafana & MongoDB dashboard for throughput/latency.
- Overhauled UVM testbenches to generate realistic constrained-random traffic; improved simulation test coverage.
- Scaled simulation runs to 1M+ transactions per DUT port, enabling early detection of critical design bottlenecks.
- Collaborated with Verification Lead to investigate detected issues and document them for RTL owners.

Undergraduate Course Assistant — Jan 2023 – Aug. 2023
University of Illinois Urbana-Champaign — *Champaign, IL*

- Held weekly office hours to mentor and support roughly 150 students in ECE385: Digital Systems Laboratory.
- Guided students through lab assignments, including the design and implementation of a multicycle RISC processor datapath, a VGA text mode controller, and an SoC platform using the Nios II soft CPU core.

Projects

RV32IM Out-of-Order Superscalar Processor | *SystemVerilog, VCS, Verdi* — Oct. 2025 – Present

- Designed and implemented an RV32IM out-of-order processor featuring a GSHARE branch predictor, pipelined L1 cache, early branch recovery with checkpointing, and next-line prefetching.
- Achieved ∼ 1.64 performance improvement over baseline and ∼ 0.8 IPC on CoreMark.
- Currently developing advanced features including a TAGE branch predictor and 2-way superscalar pipeline.

tageBuilder - TAGE branch predictor performance modelling | *Python* — Nov. 2024 – Feb. 2025

- Implemented a parameterized TAGE-like branch predictor model to aid rapid microarchitectural decision.
- Processed 200+ BT9 traces with batch runs, emitted live MPKI, accuracy, SRAM size, misprediction stats, branch-class analysis with plots + CSV export.
- Utilizing the tool yielded a predictor configuration that scores accuracy of 95% and MPKI of 1-3 for majority of traces within the CBP-5 dataset.

Unix-like Operating System | *C, QEMU* — Apr. 2023 – May 2023

- Collaborated within a team of four to design and implement a UNIX-like operating system for an x86 uniprocessor system. Implemented as part of ECE 391: Computer Systems Programming.
- Developed key OS features including a read-only filesystem, multithreading, virtual memory management, interrupt handling, device drivers, and support for system calls (e.g., execute, read, write).

Technical Skills

HDLs: SystemVerilog, Verilog
Programming & Tools: Python, C/C++, Linux, VCS, Verdi, ModelSim, Quartus, Grafana, MongoDB